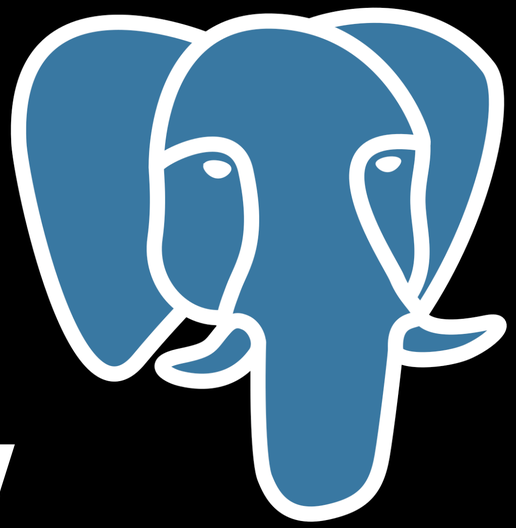


Half a dozen exciting new PostgreSQL features

PGDU.org 2025 Sydney
Thomas Munro — Open source developer at Microsoft

PostgreSQL 18 is out!

507 people are listed as developers for this release:

Abhishek Chanda
Adam Guo
Adam Rauch
Aidar Imamov
Ajin Cherian
Alastair Turner
Alec Cozens
Aleksander Alekseev
Alena Rybakina
Alex Friedman
Alex Richman
Alexander Alehin
Alexander Borisov
Alexander Korotkov
Alexander Kozhemyakin
Alexander Kukushkin
Alexander Kuzmenkov
Alexander Kuznetsov
Alexander Lakhin
Alexander Pyhalov
Alexandra Wang
Alexey Dvoichenkov
Alexey Makhmutov
Alexey Shishkin
Ali Akbar
Álvaro Herrera
Álvaro Mongil
Amit Kapila
Amit Langote
Amul Sul
Andreas Karlsson
Andreas Scherbaum
Andreas Ulbrich
Andrei Lepikhov
Andres Freund
Andrew
Andrew Bille
Andrew Dunstan
Andrew Jackson
Andrew Kane
Andrew Watkins
Andrey Borodin
Andrey Chudnovsky
Andrey Rachitskiy
Andrey Rudometov
Andy Alsup
Andy Fan
Anthonin Bonnefoy
Anthony Hsu
Anthony Leung
Anton Melnikov
Anton Voloshin
Antonin Houska
Antti Lampinen
Arseniy Mukin
Artur Zakirov
Arun Thirupathi
Ashutosh Bapat
Asphator
Atsushi Torikoshi
Avi Weinberg
Aya Iwata
Ayush Tiwari
Ayush Vatsa
Bastien Roucariès
Ben Peachey Higdon
Benoit Lobréau
Bernd Helmle
Bernd Reiß
Bernhard Wiedemann
Bertrand Drouvot
Bertrand Mamasam
Bharath Rupireddy
Bogdan Grigorenko
Boyu Yang
Braulio Fdo Gonzalez
Bruce Momjian
Bykov Ivan
Cameron Vogt
Cary Huang
Cédric Villemain
Cees van Zeeland
ChangAo Chen
Chao Li
Chapman Flack
Charles Samborski
Chengwen Wu
Chengxi Sun
Chiranmoy Bhattacharya
Chris Gooch
Christian Charukiewicz
Christoph Berg
Christophe Courtois
Christopher Inokuchi
Clemens Ruck
Corey Huinker
Craig Milhiser
Crisp Lee
Dagfinn Ilmari Mannsåker
Daniel Elishakov
Daniel Gustafsson
Daniel Vérité
Daniel Westermann
Daniele Varrazzo
Daniil Davydov
Daria Shanina
Dave Cramer
Dave Page
David Benjamin
David Christensen
David Fiedler
David G. Johnston
David Geier
David Rowley
David Steele
David Wheeler
David Zhang
Davinder Singh
Dean Rasheed
Devanga Susmitha
Devrim Gündüz
Dian Fay
Dilip Kumar
Dimitrios Apostolou
Dipesh Dhameliya
Dmitrii Bondar
Dmitry Dolgov
Dmitry Koval
Dmitry Kovalenko
Dmitry Yurichev
Dominique Devienne
Donghang Lin
Dorjpalam Batbaatar
Drew Callahan
Duncan Sands
Dwayne Towell
Dzmitry Jachnik
Egor Chindyaskin
Egor Rogov
Emanuel Ionescu
Emanuele Musella
Emre Hasegeli
Eric Cyr
Erica Zhang
Erik Nordström
Erik Rijkers
Erik Wienhold
Erki Eessaar
Ethan Mertz
Etienne LAFARGE
Etsuro Fujita
Euler Taveira
Evan Si
Evgeniy Gorbanev
Fabio R. Sluzala
Fabrízio de Royes Mello
Feike Steenbergen
Feliphe Pozzer
Felix
Fire Emerald
Florents Tselai
Francesco Degrassi
Frank Streitzig
Frédéric Yhuel
Fredrik Widlert
Gabriele Bartolini
Gavin Panella
Geoff Winkless
George MacKerron
Gilles Darold
Grant Gryczan
Greg Burd
Greg Sabino Mullane
Greg Stark
Grigory Kryachko
Guillaume Lelarge
Gunnar Morling
Gunnar Wagner
Gurjeet Singh
Haifang Wang
Hajime Matsunaga
Hamid Akhtar
Hannu Krosing
Hari Krishna Sunder
Haruka Takatsuka
Hayato Kuroda
Heikki Linnakangas
Hironobu Suzuki
Holger Jakobs
Hubert Lubaczewski
Hugo Dubois
Hugo Zhang
Hunaid Sohail
Hywel Carver
Ian Barwick
Ibrar Ahmed
Igor Gnatyuk
Igor Korot
Ilia Evdokimov
Ilya Gladyshev
Ilyasov Ian
Imran Zaheer
Isaac Morland
Israel Barth Rubio
Ivan Kush
Jacob Brazeal
Jacob Champion
Jaime Casanova
Jakob Egger
Jakub Wartak
James Coleman
James Hunter
Jan Behrens
Japin Li
Jason Smith
Jayesh Dehankar
Jeevan Chalke
Jeff Davis
Jehan-Guillaume de Rorthais
Jelte Fennema-Nio
Jian He
Jianghua Yang
Jiao Shuntian
Jim Jones
Jim Nasby
Jingtang Zhang
Jingzhou Fu
Joe Conway
Joel Jacobson
John Hutchins
John Naylor
Jonathan Katz
Jorge Solórzano
José Villanova
Josef Šimánek
Joseph Koshakow
Julien Rouhaud
Junwang Zhao
Justin Pryzby
Kaido Vaikla
Kaimeh
Karina Litskevich
Karthik S
Kartyshov Ivan
Kashif Zeeshan
Keisuke Kuroda
Kevin Hale Boyes
Kevin K Biju
Kirill Reshke
Kirill Zdornyy
Koen De Groote
Koichi Suzuki
Koki Nakamura
Konstantin Knizhnik
Kouhei Sutou
Kuntal Ghosh
Kyotaro Horiguchi
Lakshmi Narayana Velayudam
Lars Kanis
Laurence Parry
Laurenz Albe
Lele Gaifax
Li Yong
Lilian Ontowhee
Lingbin Meng
Luboslav Špilák
Luca Vallisa
Lukas Fittl
Maciek Sakrejda
Magnus Hagander
Mahendra Singh Thalor
Mahendrakar Srinivasarao
Maiquel Grassi
Maksim Korotkov
Maksim Melnikov
Man Zeng
Marat Buharov
Marc Balmer
Marco Nenciarini
Marcos Pegoraro
Marina Polyakova
Mark Callaghan
Mark Dilger
Marlene Brandstaetter
Marlene Reiterer
Martin Rakhmanov
Masahiko Sawada
Masahiro Ikeda
Masao Fujii
Mason Mackaman
Mat Arye
Matheus Alcantara
Mats Kindahl
Matthew Gabeler-Lee
Matthew Kim
Matthew Sterrett
Matthew Woodcraft
Matthias van de Meent
Matthieu Denais
Maurizio Boriani
Max Johnson
Max Madden
Maxim Boguk
Maxim Orlov
Maximilian Chrzan
Melanie Plageman
Melih Mutlu
Mert Alev
Michael Banck
Michael Bondarenko
Michael Christofides
Michael Guissine
Michael Harris
Michaël Paquier
Michail Nikolaev
Michal Kleczek
Michel Pelletier
Mikaël Gourlaouen
Mikhail Gribkov
Mikhail Kot
Milosz Chmura
Muralikrishna Bandaru
Murat Efendioglu
Mutaamba Maasha
Naeem Akhter
Nat Makarevitch
Nathan Bossart
Navneet Kumar
Nazir Bilal Yavuz
Neil Conway
Niccolò Fei
Nick Davies
Nicolas Maus
Niek Brasa
Nikhil Raj
Nikita
Nikita Kalinin
Nikita Malakhov
Nikolay Samokhvalov
Nikolay Shaplov
Nisha Moond
Nitin Jadhav
Nitin Motiani
Noah Misch
Noboru Saito
Noriyoshi Shinoda
Ole Peder Brandtzæg
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Oleg Tselebrovskiy
Olleg Samoylov
Onder Kalaci
Ondrej Navratil
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Paul Ramsey
Pavel Borisov
Pavel Luzanov
Pavel Nekrasov
Pavel Stehule
Peter Eisentraut
Peter Geoghegan
Peter Mittere
Peter Smith
Phil Eaton
Philipp Salvisberg
Philippe Beaudoin
Pierre Giraud
Pixian Shi
Polina Bungina
Przemyslaw Sztoch
Quynh Tran
Rafia Sabih
Raghuveer Devulapalli
Rahila Syed
Rama Malladi
Ran Benita
Ranier Vilela
Renan Alves Fonseca
Richard Guo
Richard Neill
Rintaro Ikeda
Robert Haas
Robert Treat
Robins Tharakan
Roman Zharkov
Ronald Cruz
Ronan Dunklau
Rui Zhao
Rushabh Lathia
Rustam Allakov
Ryo Kanbayashi
Ryohei Takahashi
RyotaK
Sagar Dilip Shedge
Salvatore Dipietro
Sam Gabrielsson
Sam James
Sameer Kumar
Sami Imseih
Samuel Thibault
Satyanarayana Narlapuram
Sebastian Skalacki
Senglee Choi
Sergei Kornilov
Sergey Belyashov
Sergey Dudoladov
Sergey Prokhorenko
Sergey Sargsyan
Sergey Soloviev
Sergey Tatarintsev
Shaik Mohammad Mujeeb
Shawn McCoy
Shenhao Wang
Shihao Zhong
Shinya Kato
Shlok Kyal
Shubham Khanna
Shveta Malik
Simon Riggs
Smolkin Grigory
Sofia Kopikova
Song Hongyu
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Stan Hu
Stepan Neretin
Stephen Fewer
Stephen Frost
Steve Chavez
Steven Niu
Suraj Kharage
Sven Klemm
Takamichi Osumi
Takeshi Ideriha
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Ted Yu
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Xuneng Zhou
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Yaroslav Syrytsia
Yasir Hussain
Yasuo Honda
Yogesh Sharma
Yonghao Lee
Yoran Heling
Yu Liang
Yugo Nagata
Yuhang Qiu
Yuki Seino
Yura Sokolov
Yurii Rashkovskii
Yushi Ogiwara
Yusuke Sugie
Yuta Katsuragi
Yuto Sasaki
Yuuki Fujii
Yuya Watari
Zane Duffield
Zeyuan Hu
Zhang Mingli
Zhihong Yu
Zhijie Hou
Zsolt Parragi
•

Talk

- Part I: Themes in recent releases
- Part II: Three selected user-facing improvements
- Part III: Three selected internal improvements

Part I: Themes in recent releases

The SQL standard

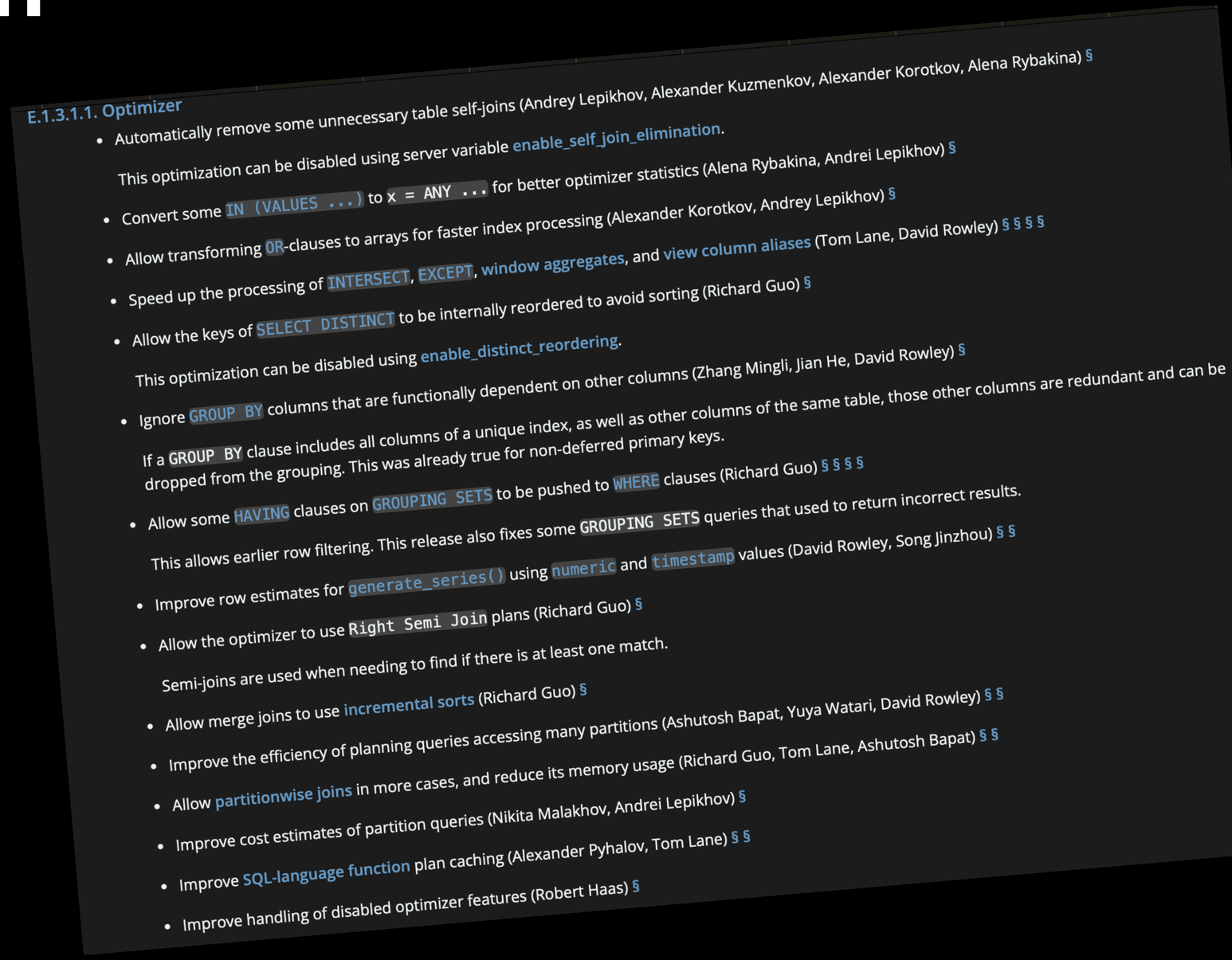
- MERGE statements — v15
- SQL/JSON — standardisation of JSON v12, v16, v17
- Elements of temporal support, small steps — v18
- Much more

- WIP: SQL/PGQ for graphs (SQL:2023)
- Opportunities: MATCH_RECOGNIZE, DECFLOAT

Planner evolution

(As always)

- Giving the planner more ways to consider executing your queries
- Being a heuristic system, new plans can sometimes turn out to be slower for a given query, requiring investigation



E.1.3.1.1. Optimizer

- Automatically remove some unnecessary table self-joins (Andrey Lepikhov, Alexander Kuzmenkov, Alexander Korotkov, Alena Rybakina) §

  This optimization can be disabled using server variable enable_self_join_elimination.

- Convert some IN (VALUES ...) to x = ANY ... for better optimizer statistics (Alena Rybakina, Andrei Lepikhov) §
- Allow transforming OR-clauses to arrays for faster index processing (Alexander Korotkov, Andrey Lepikhov) §
- Speed up the processing of INTERSECT, EXCEPT, window aggregates, and view column aliases (Tom Lane, David Rowley) § § § §
- Allow the keys of SELECT DISTINCT to be internally reordered to avoid sorting (Richard Guo) §

  This optimization can be disabled using enable_distinct_reordering.

- Ignore GROUP BY columns that are functionally dependent on other columns (Zhang Mingli, Jian He, David Rowley) §

  If a GROUP BY clause includes all columns of a unique index, as well as other columns of the same table, those other columns are redundant and can be dropped from the grouping. This was already true for non-deferred primary keys.

- Allow some HAVING clauses on GROUPING SETS to be pushed to WHERE clauses (Richard Guo) § § § §

  This allows earlier row filtering. This release also fixes some GROUPING SETS queries that used to return incorrect results.

- Improve row estimates for generate_series() using numeric and timestamp values (David Rowley, Song Jinzhou) § §
- Allow the optimizer to use Right Semi Join plans (Richard Guo) §

  Semi-joins are used when needing to find if there is at least one match.

- Allow merge joins to use incremental sorts (Richard Guo) §
- Improve the efficiency of planning queries accessing many partitions (Ashutosh Bapat, Yuya Watari, David Rowley) § §
- Allow partitionwise joins in more cases, and reduce its memory usage (Richard Guo, Tom Lane, Ashutosh Bapat) § §
- Improve cost estimates of partition queries (Nikita Malakhov, Andrei Lepikhov) §
- Improve SQL-language function plan caching (Alexander Pyhalov, Tom Lane) § §
- Improve handling of disabled optimizer features (Robert Haas) §

Indexes

- Btree — workhorse index AM
  - Parallel index build — v11
  - Duplicate compression — v12
  - Efficient IN scan — v17
  - Skip scan — v18
  - Faster build by skipping buffer pool and writing directly to disk
- Equivalent GIN, Gist, BRIN, Hash index performance improvements tend to follow later (where appropriate)

Observability

- Overhauled cumulative statistics subsystem
- New I/O statistics views: pg_stat_io — v16
- EXPLAIN improved to show information about buffers, WAL, caching, etc

I/O performance

- V16: Allocating disk space in bulk during large data insertions (XFS, EXT4)
- Baby steps in optimal data exchange between storage and memory
  - V16: Experimental direct I/O mode to skip kernel page cache
  - V17: Streaming I/O, combined I/O; limited usage but it's fast
  - V18: Asynchronous I/O
- Various ways to take advantage of reflinks (XFS, EXT4, …), block sharing (XFS, BTRFS, ZFS) and copy push-down (network file systems, …), CREATE DATABASE, pg_upgrade, pg_combinebackup (—> see later)

Recovery and replication

- Physical replication and crash recovery
  - V14 — performance improvements
  - V15 — basic prefetching of data pages during crash recovery and binary replication (work ongoing)
- Logical replication
  - Parallelisation, completeness, failover, maintenance improvements

Modernisation

- Migration from GNU configure + make build system to meson build system, making building testing faster and the same across Unix and Windows systems
- Dropping support for OSes, compilers, library versions that have reached the end of the line, don’t have active maintainers, aren’t keeping up with the modern standards we need for our plans
- V19: Moving to C11, requiring recent POSIX features and recent Windows versions
- V15: CI

Created 1 day ago

Finished in 18:42

[CF 5790] v16 - Add --continue-on-abort option to pgbench

Commit 51d917f on branch cf/5790.

TASKS (10)

HOOKS (0)

00:44	SanityCheck
07:56	FreeBSD - Meson
04:58	NetBSD - Meson
09:31	OpenBSD - Meson
12:37	Linux - Debian Bookworm - Autoconf
14:57	Linux - Debian Bookworm - Meson
06:16	macOS - Sonoma - Meson
17:58	Windows - Server 2019, VS 2019 - Meson & ninja
17:21	Windows - Server 2019, MinGW64 - Meson

Low level optimisation

“Mechanical sympathy”

- Removing false sharing, unnecessary writes, lock contention
- Vectorising, autovectorising,
- Specialising in C
- Specialising with JIT
- Detecting and using special hardware instructions
- Batching
- Tuning layout
- Baby steps in understanding NUMA

Many more areas

- Performance of partitioned tables
- Better support for collations: fexibility and stability
- Certainly many things I’m forgetting!

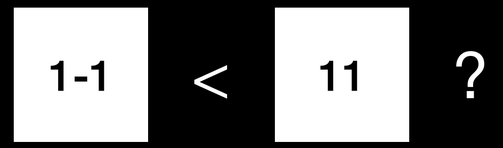
Part II: Three selected user-facing improvements

SQL/JSON

- Alignment with SQL:2016, SQL:2023 standardised JSON support
- JSON_TABLE() — query JSON like a table in a query
- JSON_QUERY() — standardised JPath queries

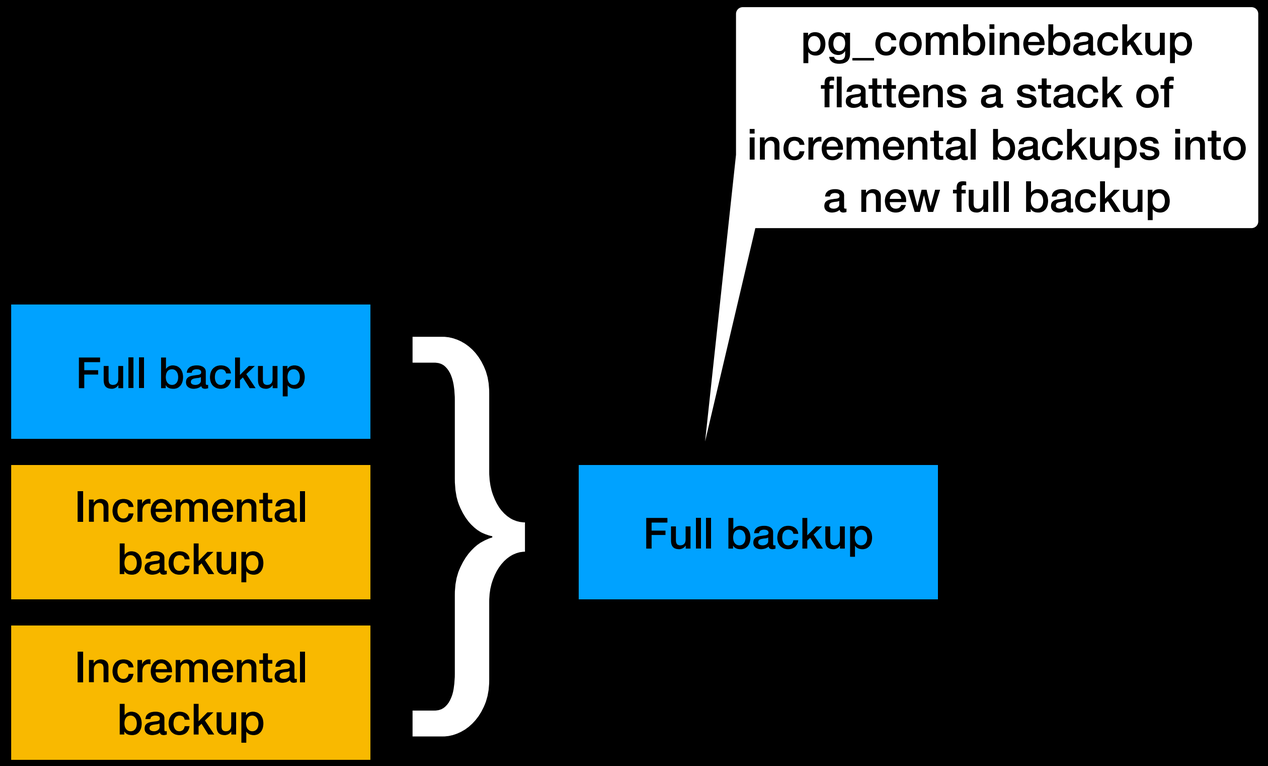
New options for locales

- New built-in locale provider allowing for fast Unicode text without unstable sort order determined by system libraries
- Changes to glibc, ICU definitions can cause indexes to become corrupt
- Only C was completely stable in the past
- Historically even C.UTF8 had surprising instability
- Many data sets don’t need cultural text sorting rules, and users may not realize they are being used until something breaks



Incremental backups — v17

pg_basebackup, pg_combinebackup



Combined backup can even share disk blocks on COW

Part III: Internal improvements

Shared memory cumulative stats system

pg_stat_* views, pg_statio_* views, etc

V14

```
/path/to/bin/postgres -D pgdata
├─ postgres: checkpointer
├─ postgres: background writer
├─ postgres: walwriter
├─ postgres: autovacuum launcher
├─ postgres: logical replication launcher
├─ postgres: stats collector
├─ postgres: myuser mydb [local] idle
├─ postgres: myuser mydb [local] idle
└─ postgres: myuser mydb [local] idle
```

Receives updates via a UDP packet, writes total set of statistics to a file

Backends would would read a copy of the statistics file into memory

V15

```
/path/to/bin/postgres -D pgdata
├─ postgres: checkpointer
├─ postgres: background writer
├─ postgres: walwriter
├─ postgres: autovacuum launcher
├─ postgres: logical replication launcher
├─ postgres: myuser mydb [local] idle
├─ postgres: myuser mydb [local] idle
└─ postgres: myuser mydb [local] idle
```

Backends now share a single set of statistics in shared memory, persisted to disk periodically

Skip scan for btree indexes

```
CREATE TABLE address (id int primary key, state text, city text, …);
CREATE INDEX ON address (state, city);
```

```
SELECT … FROM address WHERE state = ? AND city = ?
```

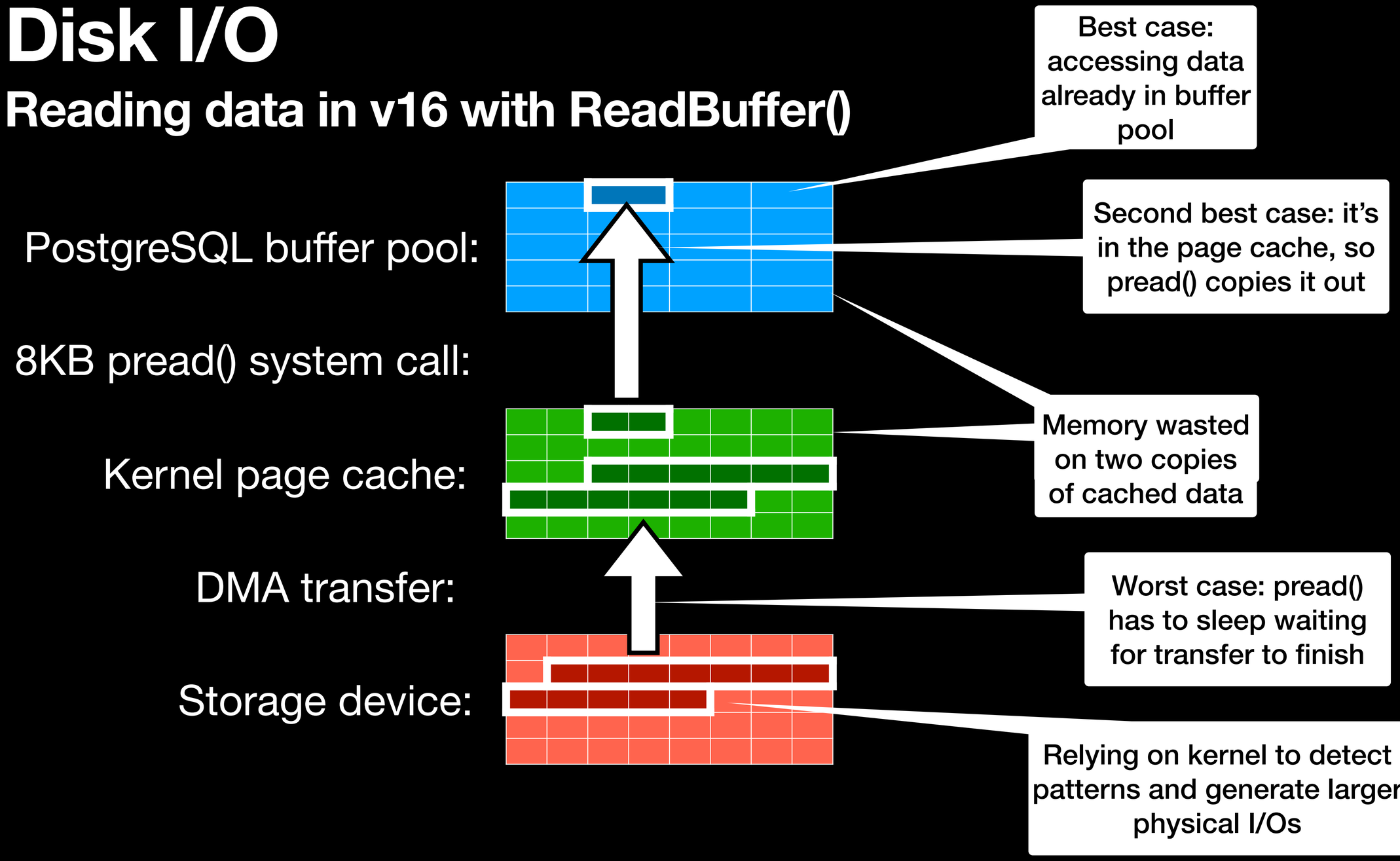
WHERE clause matches index keys

```
SELECT … FROM address WHERE city = ?
```

WHERE clause lacks initial index key, but planner considers *skipping* to each state value, which might be cheap enough for low

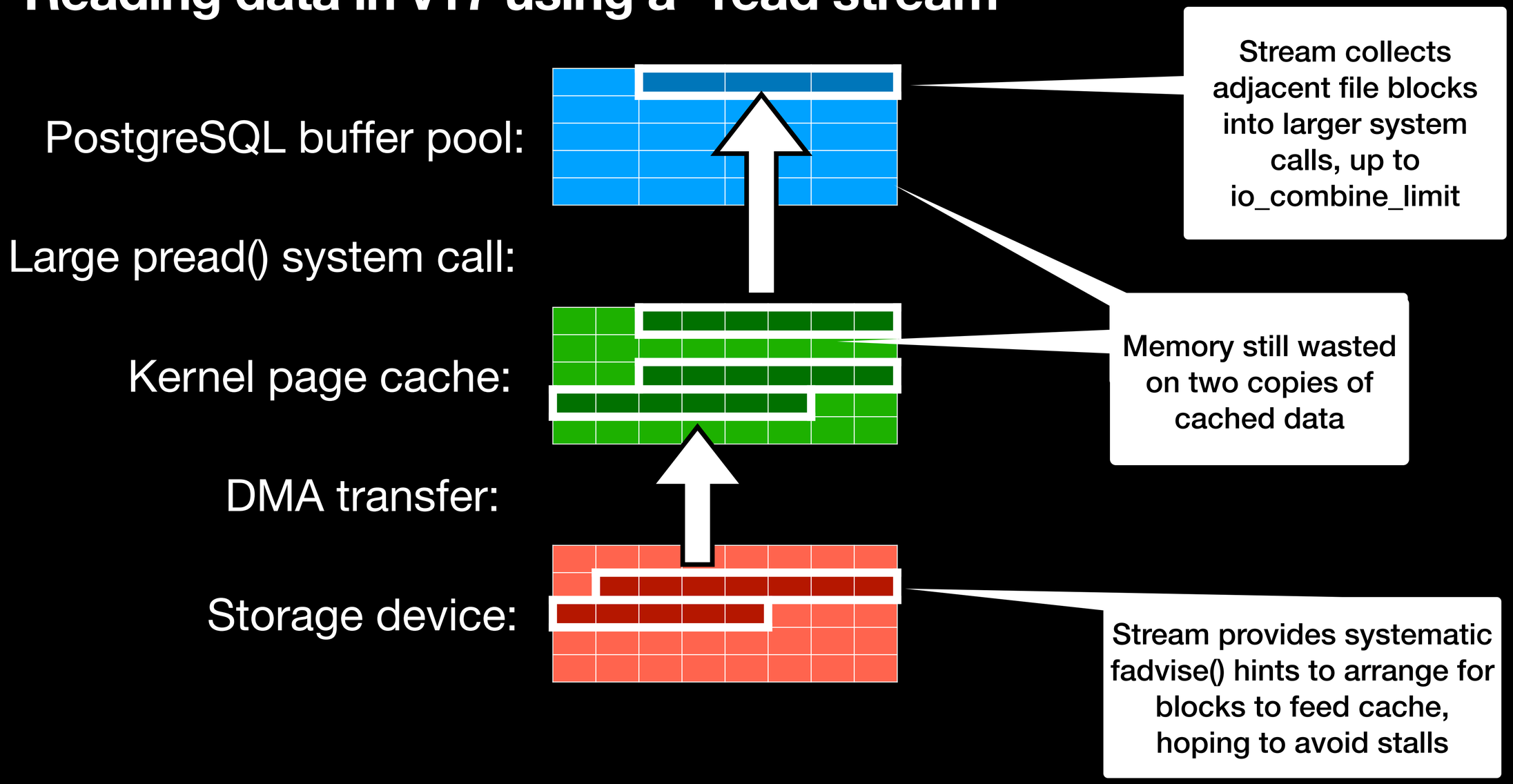
Disk I/O

Reading data in v16 with ReadBuffer()



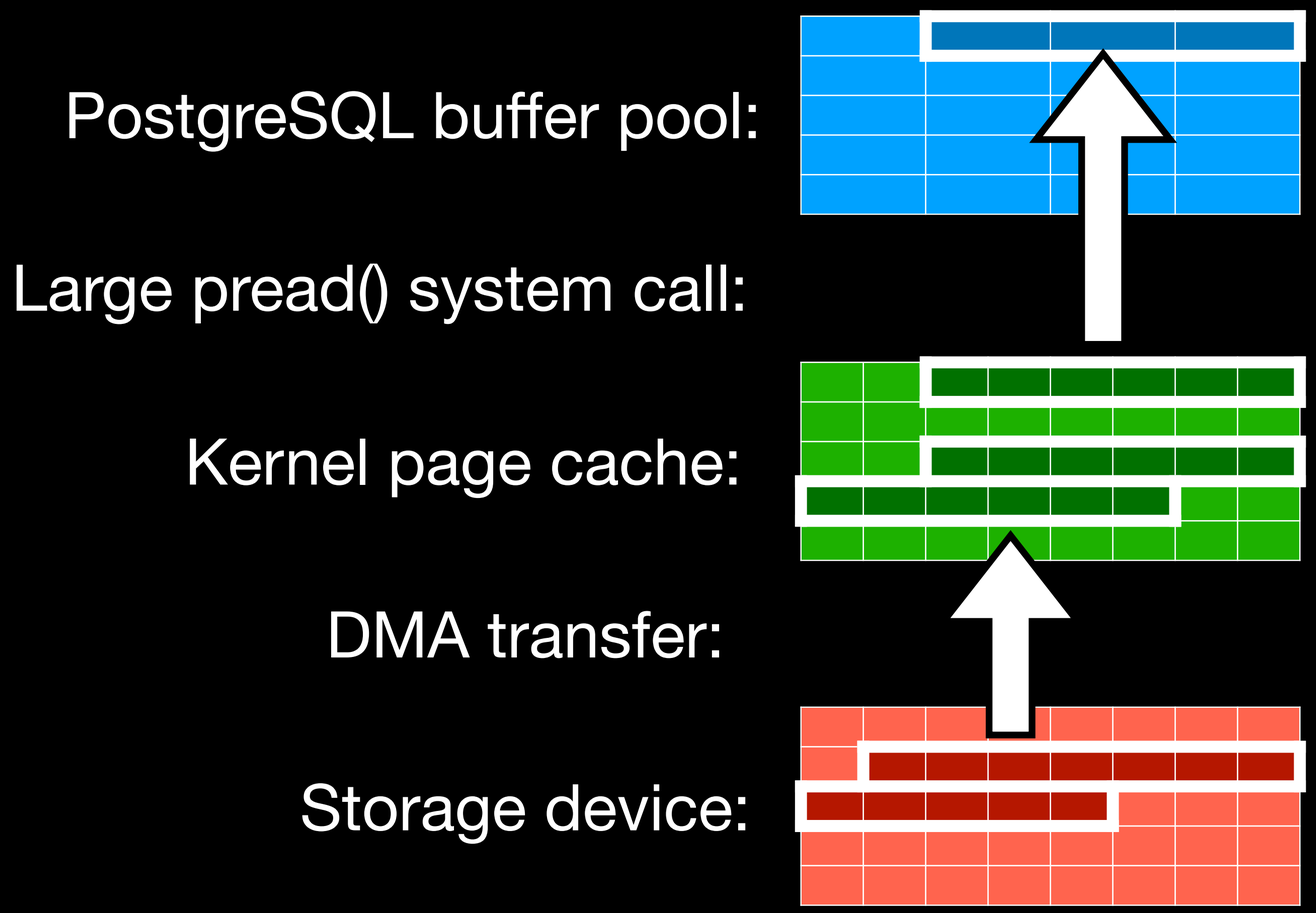
Disk I/O

Reading data in v17 using a “read stream”

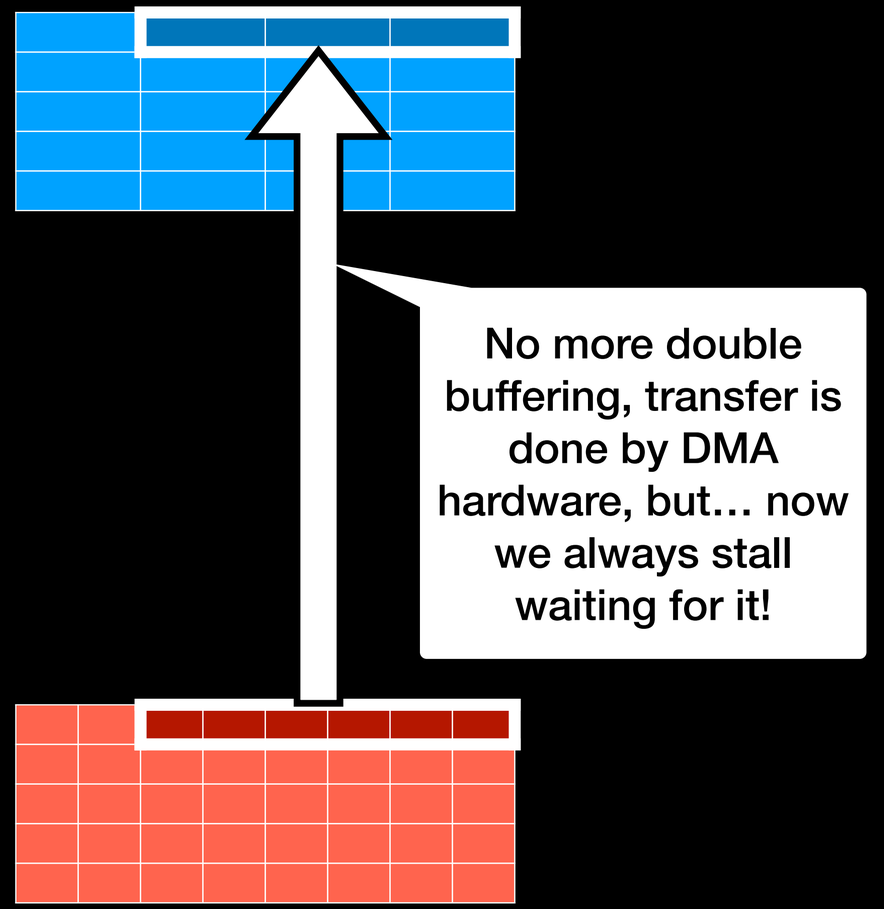


Disk I/O

Reading data in v17 using a “read stream”



With experimental direct I/O mode enabled:



Direct I/O is an optimisation AND a pessimisation! Unless…

Disk I/O

Reading data in v18 with AIO

With experimental direct I/O mode enabled:

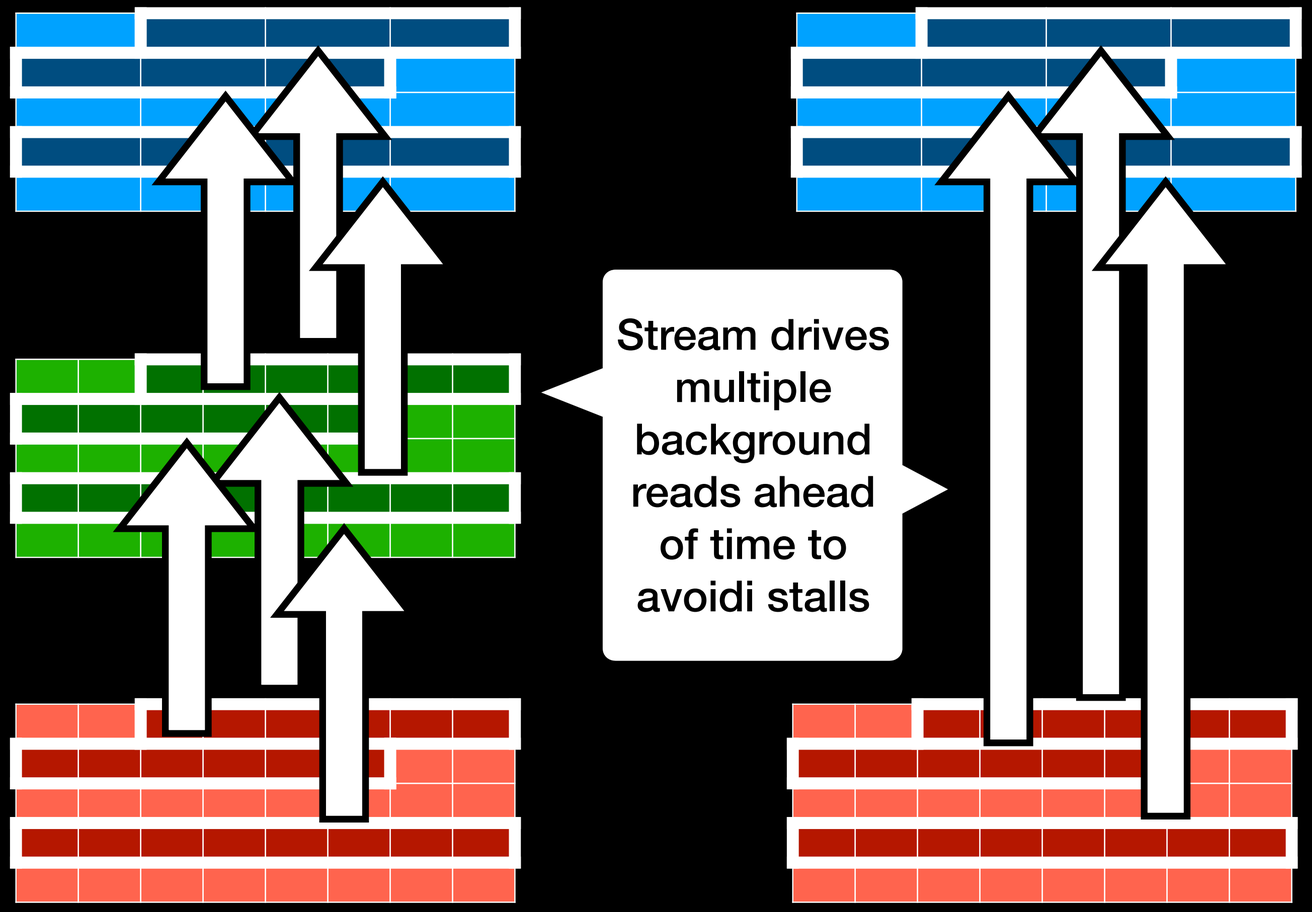
PostgreSQL buffer pool:

Large asynchronous reads:

Kernel page cache:

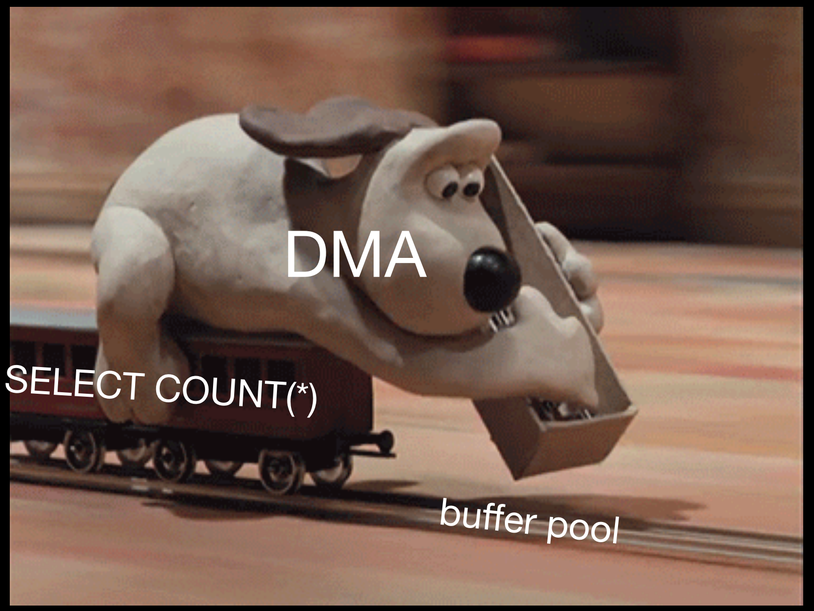
DMA transfers:

Storage device:



Disk I/O

Executor avoids stalls and copying



Video credit:

Wallace & Gromit “The Wrong Trousers”, 1993, by Aardman
Cycle loop from giphy.com

io_method=worker

```
/path/to/bin/postgres -D pgdata
 ├─ postgres: io worker worker: 0
 ├─ postgres: io worker worker: 1
 ├─ postgres: io worker worker: 2
 ├─ postgres: checkpointer
 ├─ postgres: background writer
 ├─ postgres: walwriter
 ├─ postgres: autovacuum launcher
 ├─ postgres: logical replication launcher
 └─ postgres: myuser mydb [local] SELECT
```

Running synchronous I/Os

Queues up I/O requests for workers

io_method=io_uring

```
/path/to/bin/postgres -D pgdata
 ├─ postgres: checkpointer
 ├─ postgres: background writer
 ├─ postgres: walwriter
 ├─ postgres: autovacuum launcher
 ├─ postgres: logical replication launcher
 └─ postgres: myuser mydb [local] SELECT
```

Queues up I/O requests in kernel

Disk I/O

Progress converting parts of PostgreSQL to the new I/O APIs

- V17
  - Sequential scans
  - ANALYZE
  - pg_prewarm
- V18
  - VACUUM
  - CREATE DATABASE
  - Bitmap Heap Scan
- WIP
  - Index scans (many)
  - Writes…
  - WAL…
  - Network…?

EOF